Lab 4: Auto Loader – Schema Evolution with `addNewColumns`

Lab Summary

In this lab, we test **schema evolution in Databricks Auto Loader** using the `addNewColumns` mode.

We simulate a situation where **new columns appear in incoming files** and observe how Auto Loader **automatically adds those columns to the destination Delta table** using `mergeSchema`.

This lab helps understand the difference between:

- `addNewColumns` → automatically adds new columns to the destination table
- `rescue` **mode** → stores unexpected columns in `_rescued_data`

Prerequisites

Before starting the lab, ensure the following:

- Databricks workspace is running
- Bronze volume already created
- Auto Loader raw data folder exists
- Three CSV files available:
  - **file1.csv**
  - **file2.csv**
  - **file3.csv** (contains new columns)

Directory structure:

```
bronze
 └── volumes
      └── autovol
```

```
            ├── raw
            └── destination
```

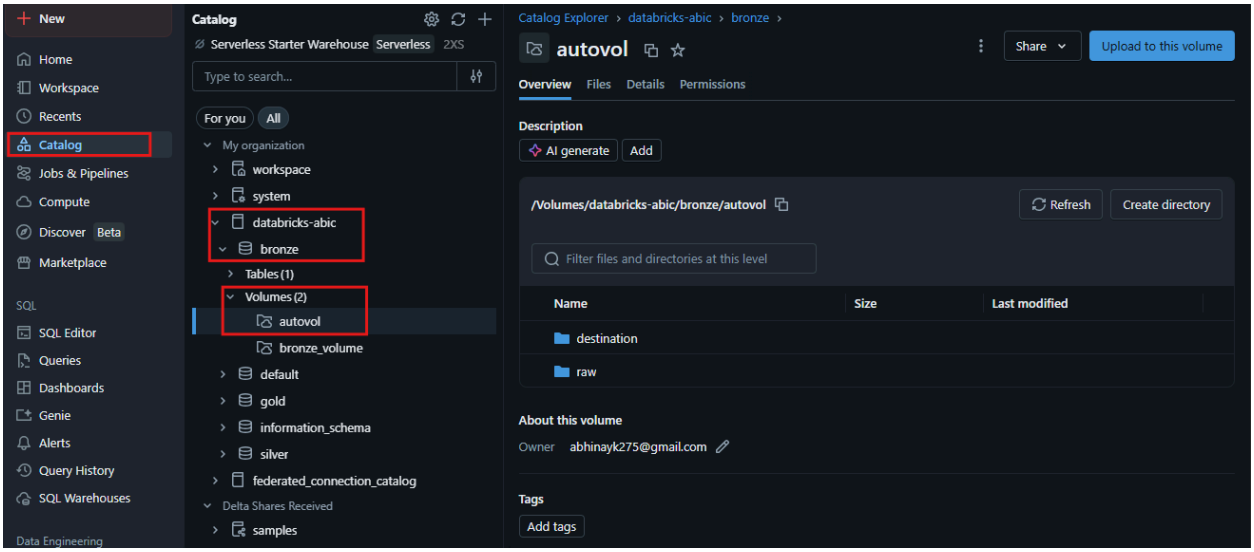
Steps to Complete Lab

Task 1: Clean Previous Data

Step 1: Navigate to Bronze Volume

Go to:

```
Databricks Workspace → Catalog → bronze → volumes → autovol
```



Step 2: Delete Existing Destination Folder

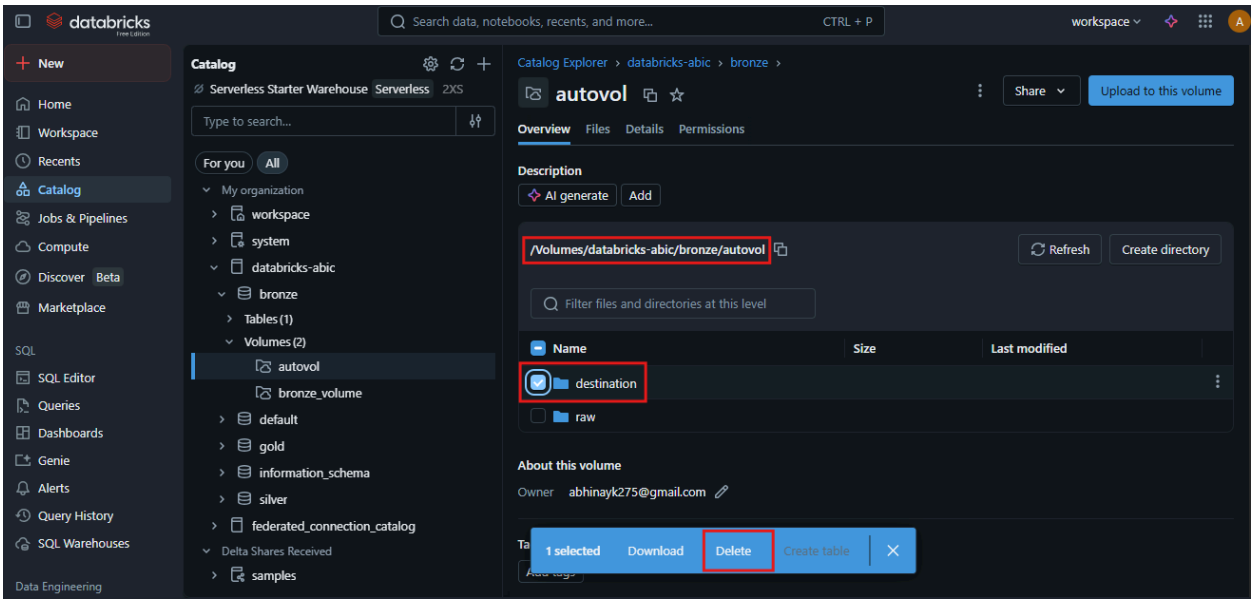
Delete the existing destination directory.

Reason:

- Previous **checkpoint logs and schema files** remain stored
- Auto Loader may reuse old schema information
- We want a **fresh pipeline execution**

Delete:

```
destination
```

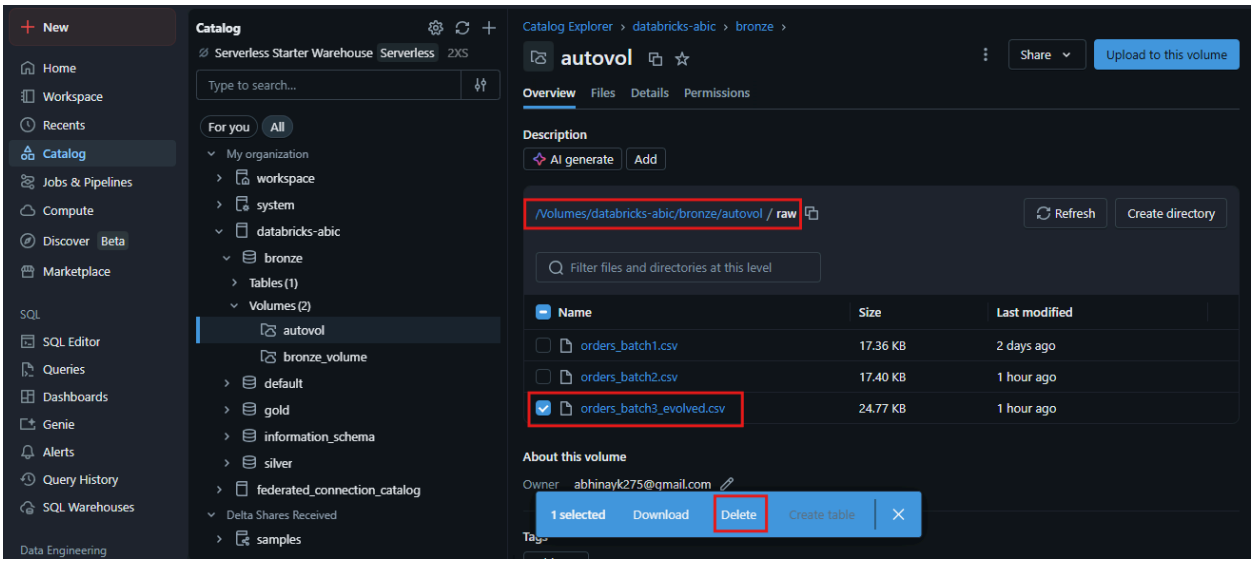


Step 3: Keep Only the First Two Files

Inside the `raw` folder ensure only:

```
file1.csv
file2.csv
```

Remove the third file for now.



Reason:

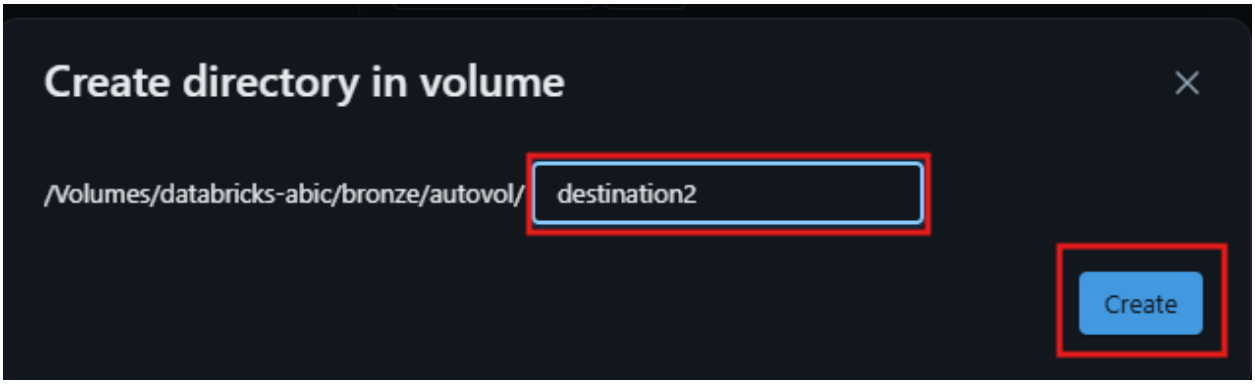
We first want Auto Loader to infer the **initial schema** from these two files.

Task 2: Create a New Destination Folder

Create a new folder to avoid any leftover metadata.

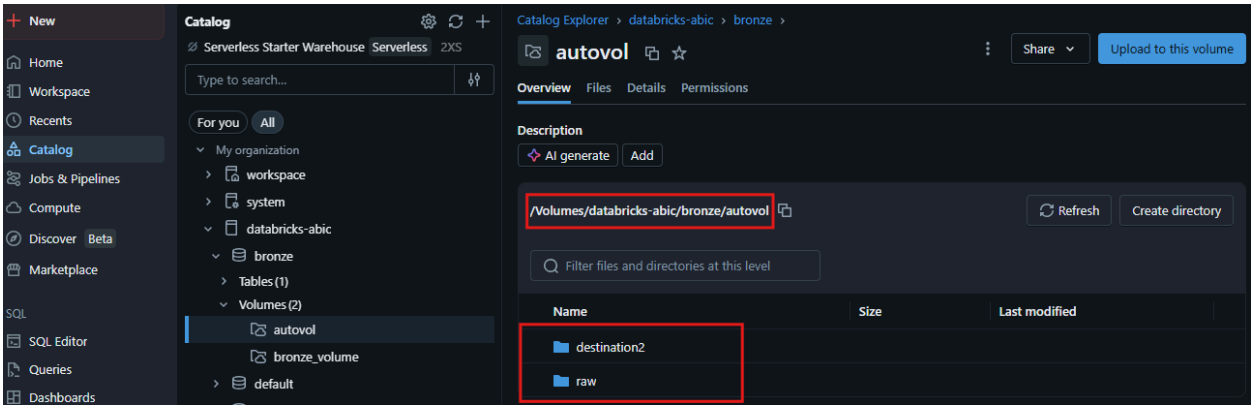
Example:

```
destination2
```



Structure:

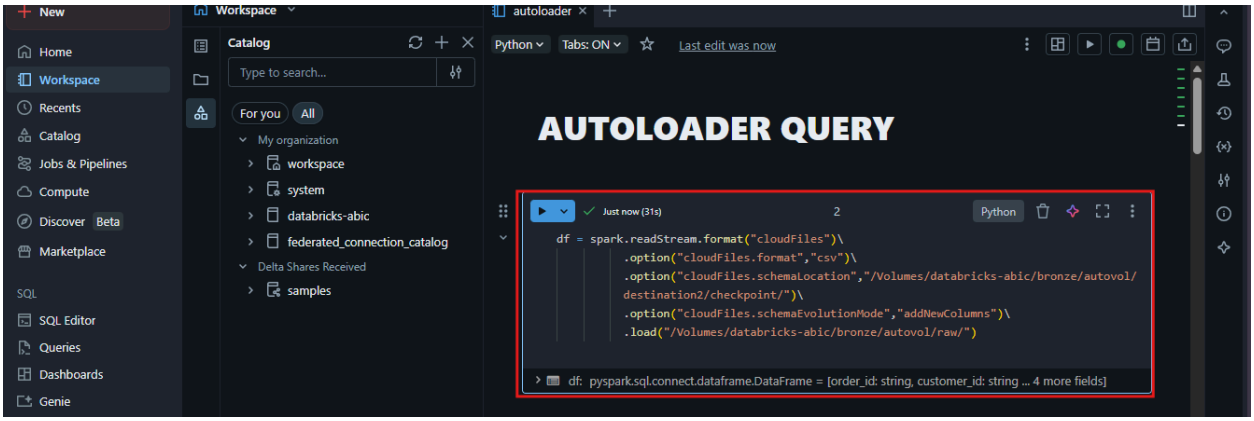
```
autovol
 ├── raw
 └── destination2
```



Task 3: Create Auto Loader Streaming Query

Run the following Auto Loader streaming read:

```
df = spark.readStream.format("cloudFiles") \
        .option("cloudFiles.format","csv") \
        .option("header","true") \
        .option("cloudFiles.schemaLocation","/Volumes/databri
cks-abic/bronze/autovol/destination2/checkpoint/") \
        .option("cloudFiles.schemaEvolutionMode","addNewColum
ns") \
        .load("/Volumes/databricks-abic/bronze/autovol/raw/")
```



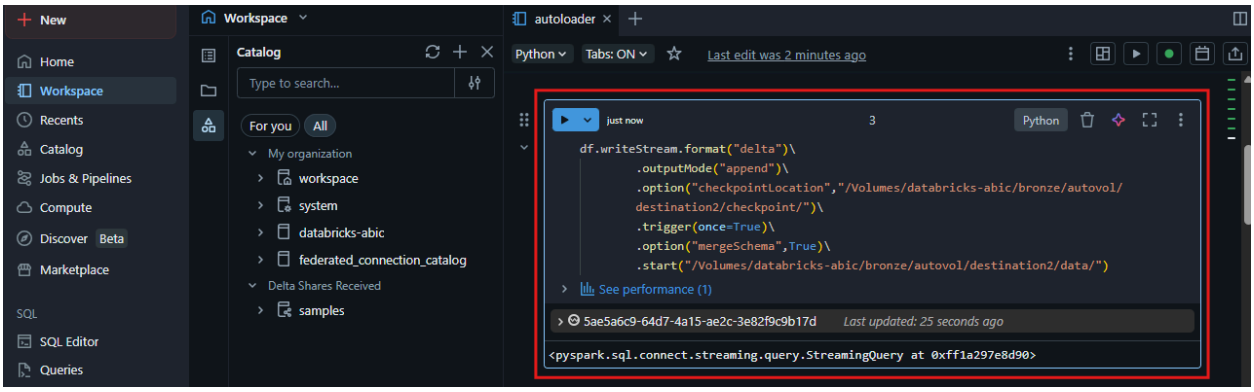
Explanation

Option	Purpose
cloudFiles.format	File format
header=true	Uses first row as column names
schemaLocation	Stores inferred schema
schemaEvolutionMode=addNewColumns	Automatically adds new columns

Task 4: Write Streaming Data to Delta Table

Now write the streaming data to the destination table.

```
df.writeStream.format("delta") \
        .outputMode("append") \
        .option("checkpointLocation","/Volumes/databricks-abic/bronze/autovol/destination2/checkpoint/") \
        .trigger(once=True) \
        .option("mergeSchema",True) \
        .start("/Volumes/databricks-abic/bronze/autovol/destination2/data/")
```



Why `mergeSchema = true`?

Because we are using:

```
schemaEvolutionMode = addNewColumns
```

When new columns appear:

- Auto Loader detects them
- `mergeSchema` allows Delta table schema to **expand automatically**

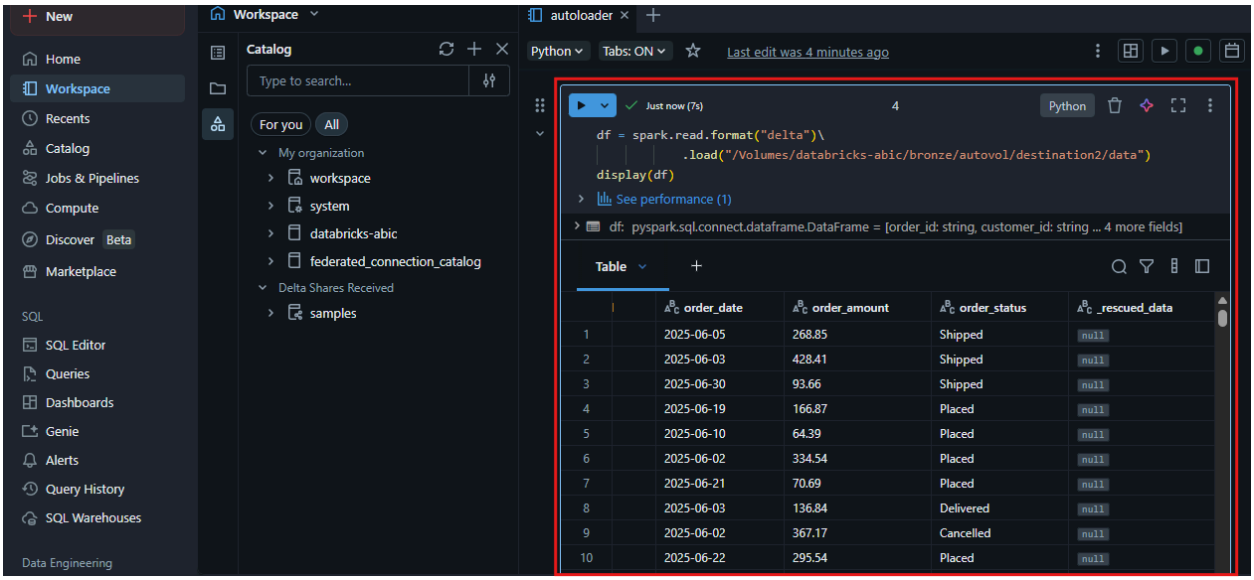
Without `mergeSchema`, the streaming job may fail when new columns arrive.

Task 5: Verify Data in Delta Table

Read the destination table.

```
df = spark.read.format("delta") \
        .load("/Volumes/databricks-abic/bronze/autovol/destin
ation2/data/")

display(df)
```



Task 6: Upload Third File (Schema Change)

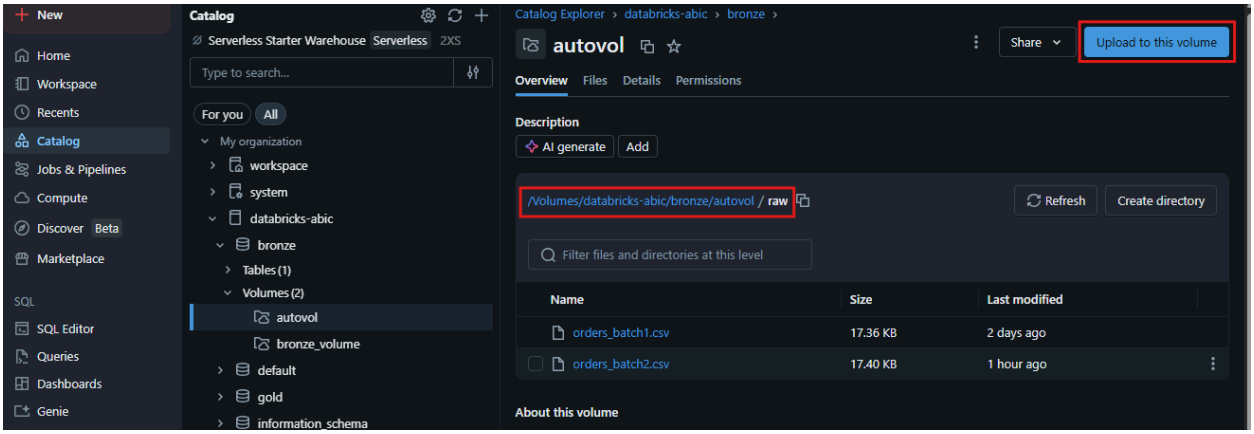
Now upload the third file which contains **new columns**.

Example new columns:

```
discount
payment_method
```

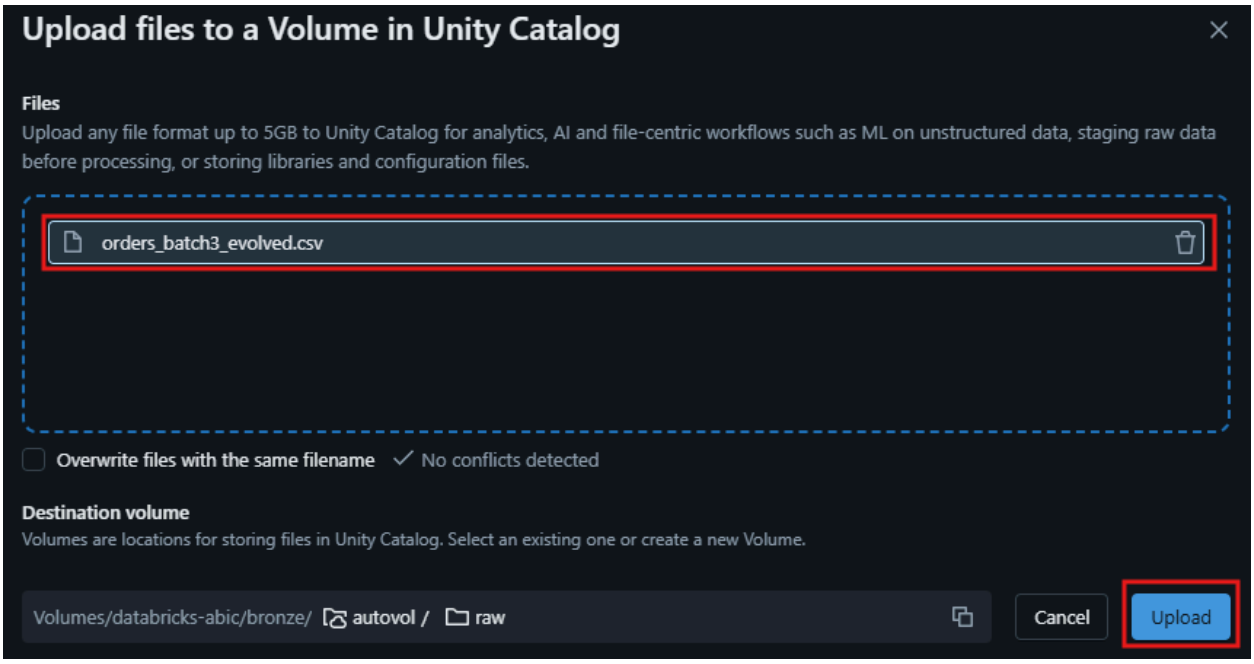
Upload:

```
bronze → volumes → autovol → raw
```



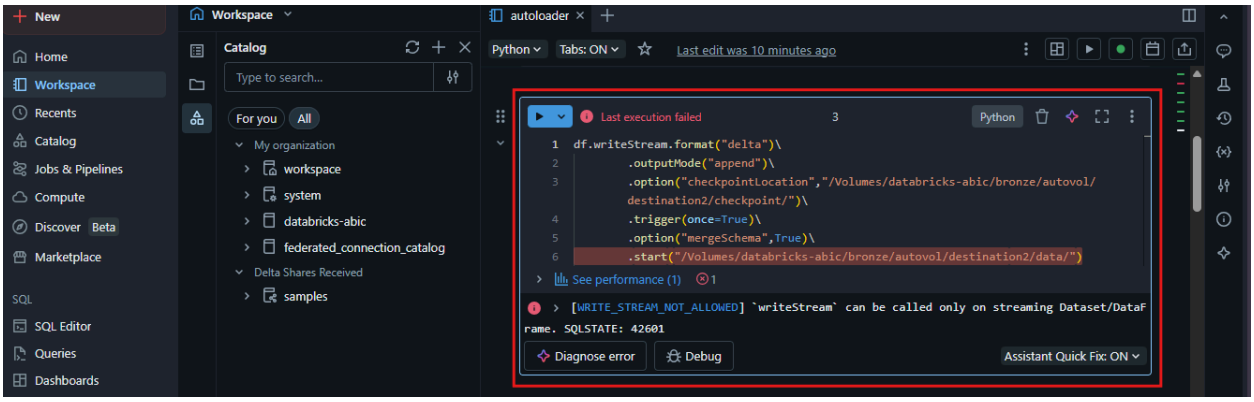
Add:

```
orders_batch3_evolved.csv
```



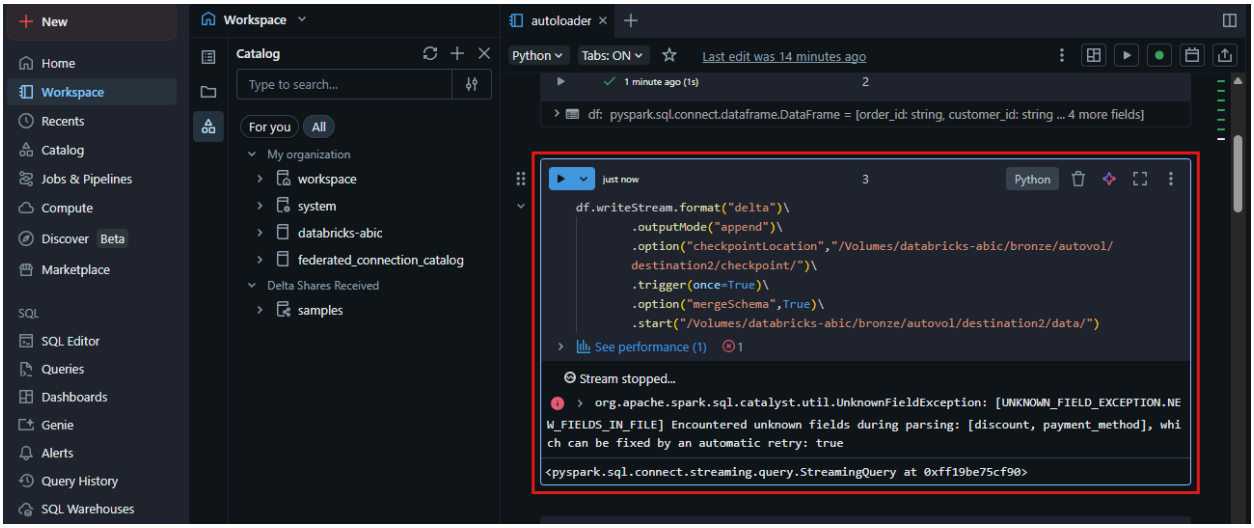
Task 7: Run the Streaming Query Again

Re-run the streaming query.



You may see an error like:

```
Unknown field exception
```

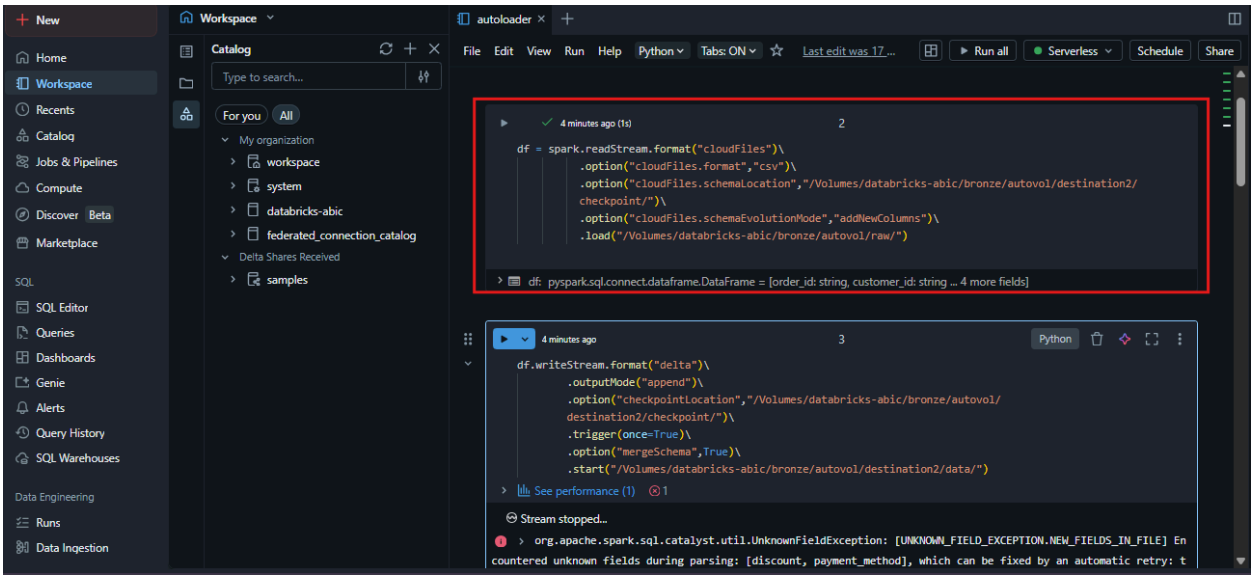


This happens because:

- Auto Loader detected a **new schema**
- The streaming query needs to **restart to reload the schema**

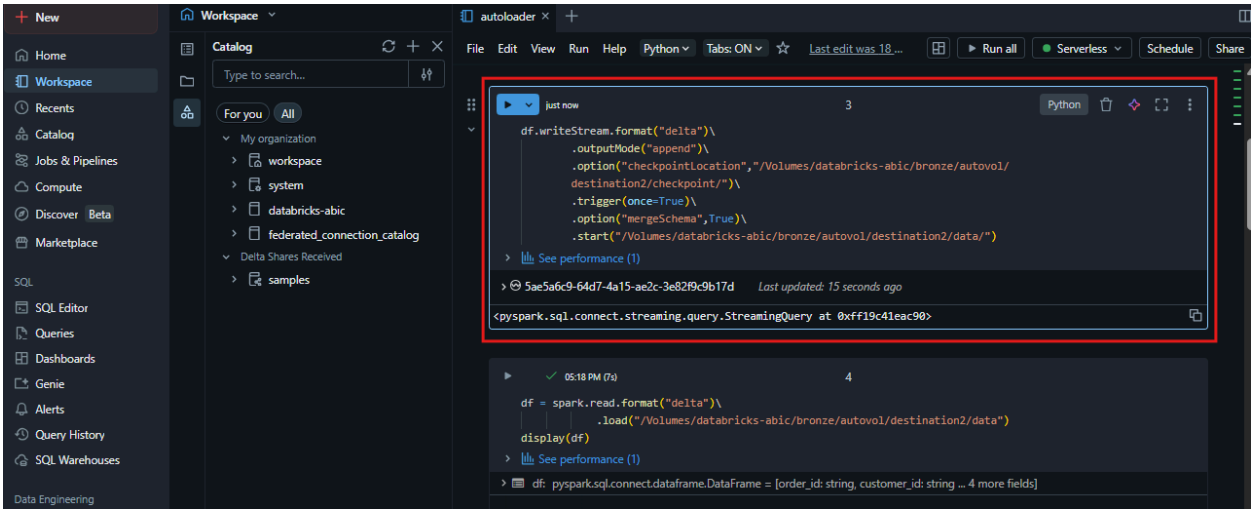
Task 8: Restart Streaming Job

Run the streaming write query again.



Now Auto Loader will:

```
Detect new columns
Update schema
Merge schema in Delta table
Write data
```



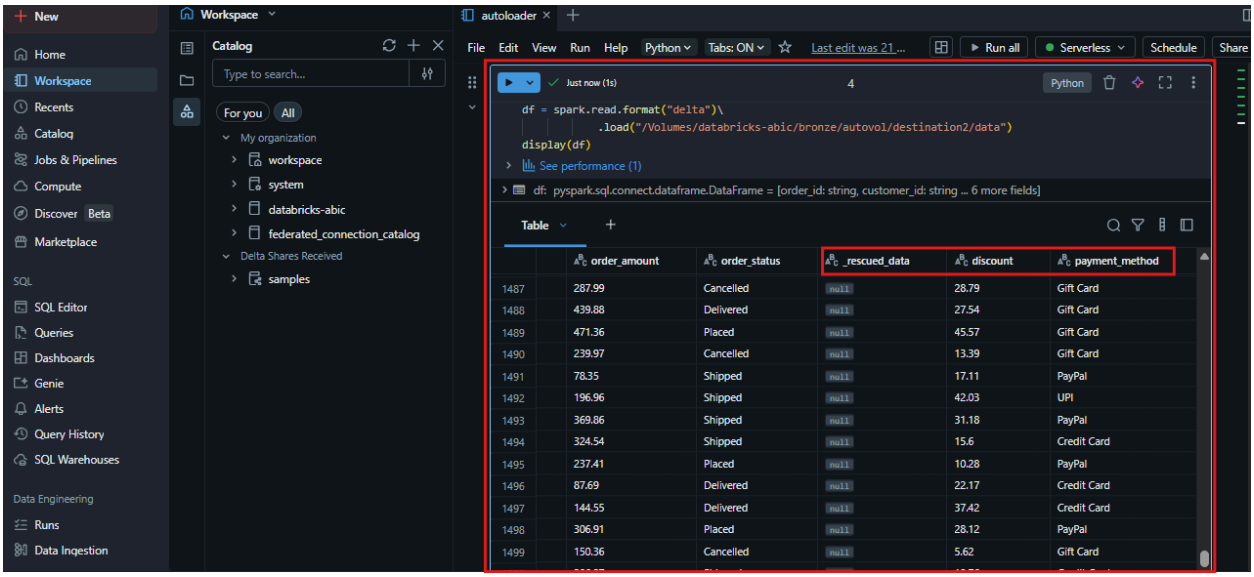
Task 9: Verify Final Data

Read the destination table again.

```
df = spark.read.format("delta") \
        .load("/Volumes/databricks-abic/bronze/autovol/destin
ation2/data/")

display(df)
```

Now you should see:



Observations:

- **New columns are added automatically**
- `_rescued_data` remains **NULL**

Reason:

```
schemaEvolutionMode = addNewColumns
```

So columns were **added instead of rescued**.

Outcome

After completing this lab you should understand:

- How **Auto Loader schema evolution works**
- Difference between **addNewColumns and rescue**
- Why **mergeSchema is required**
- How to handle schema changes in streaming pipelines.